

so expectations of layoffs add a *firing wedge* $\beta\phi$ to the effective marginal cost of labor at $t = 0$, lowering the optimal n_0 .

Closed-form comparative statics (Cobb–Douglas). With $f(n) = n^\alpha$ and $0 < \alpha < 1$, the period-0 choice under the contraction (anticipated-firing) regime satisfies

$$n_0^C = \left(\frac{\alpha A_0}{w_0 + c_h + \beta\phi} \right)^{\frac{1}{1-\alpha}}. \quad (18)$$

At $t = 1$, the firing-region choice solves

$$n_1^- = \left(\frac{\alpha A_1}{w_1 - \phi} \right)^{\frac{1}{1-\alpha}} \quad (\text{assuming } w_1 > \phi). \quad (19)$$

The contraction regime is relevant whenever

$$\frac{A_1}{w_1 - \phi} \leq \frac{A_0}{w_0 + c_h + \beta\phi}. \quad (20)$$

Differentiating (18) yields

$$\frac{\partial n_0^C}{\partial \phi} = -\frac{\beta}{1-\alpha} \cdot \frac{n_0^C}{w_0 + c_h + \beta\phi} < 0, \quad (21)$$

which captures the *option value of restraint*: hiring less today mitigates expected firing costs tomorrow.

A.5.2 Embedding the Mechanism in a Task-Based Model

To connect with canonical theories of task displacement, suppose production in each period requires a unit measure of tasks $z \in [0, 1]$. Tasks $z < I_t$ are automated (performed by capital) and $z \geq I_t$ are performed by labor; write $\Gamma_t \equiv 1 - I_t$ for the labor share of tasks. Output aggregates capital- and labor-performed tasks with CES elasticity $\sigma > 1$,

$$Y_t = A_t \left[\Gamma_t^{1/\sigma} n_t^{(\sigma-1)/\sigma} + (1 - \Gamma_t)^{1/\sigma} (A^K K)^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}, \quad (22)$$

where capital K is fixed and the constants are absorbed into Ω_t below to simplify notation. The MPL implied by (22) is

$$\frac{\partial Y_t}{\partial n_t} = A_t \left[\Gamma_t^{1/\sigma} n_t^{(\sigma-1)/\sigma} + \Omega_t \right]^{1/(\sigma-1)} \Gamma_t^{1/\sigma} n_t^{-1/\sigma}. \quad (23)$$

The firm now maximizes (16) with $A_t f(n_t)$ replaced by Y_t . The optimality conditions take exactly the same form as in the baseline:

$$\frac{\partial Y_1}{\partial n_1} = \begin{cases} w_1, & n_1 \geq n_0, \\ w_1 - \phi, & n_1 < n_0, \end{cases} \quad \frac{\partial Y_0}{\partial n_0} = \begin{cases} w_0 + c_h, & \text{no firing expected,} \\ w_0 + c_h + \beta\phi, & \text{firing anticipated.} \end{cases} \quad (24)$$

Implications. Current automation lowers Γ_0 and hence the MPL of labor, reducing n_0 . Anticipated future automation (higher I_1 and lower Γ_1) makes separations more likely at $t = 1$. When the firm expects to be in the firing regime tomorrow, the period-0 condition (24) features the same firing wedge $\beta\phi$, which depresses current hiring. Larger firing costs ϕ strengthen this effect. Consequently, in equilibrium $n_0^C < n_0^E$ (where E denotes the no-firing regime), $\partial n_0 / \partial I_0 < 0$, and $\partial n_0 / \partial (\mathbb{E}_0 I_1) < 0$.

In both formulations, the key force is the intertemporal link created by convex adjustment on the downside: the possibility of paying ϕ tomorrow raises the effective marginal cost of hiring by $\beta\phi$ today. Expectations of labor-saving progress therefore reduce current employment even before automation is realized.

A.6 Conceptual Framework: Formal Details

This appendix provides the formal derivations and proofs for the conceptual framework in Section 3.

A.6.1 Threshold Shift

After GenAI adoption with augmentation (δ_J, δ_S) , the post-AI allocation threshold $\hat{x}'$ is determined by the condition $\tilde{\gamma}_J(\hat{x}')/w_J = \tilde{\gamma}_S(\hat{x}')/w_S$, or equivalently

$$\frac{\gamma_S(\hat{x}')}{\gamma_J(\hat{x}')} = \frac{1 + \delta_J}{1 + \delta_S} \cdot \frac{w_S}{w_J} = \frac{1 + \delta_J}{1 + \delta_S} \cdot \frac{\gamma_S(\hat{x})}{\gamma_J(\hat{x})}. \quad (25)$$

When $\delta_J = \delta_S$, the right-hand side equals $\gamma_S(\hat{x})/\gamma_J(\hat{x})$, so $\hat{x}' = \hat{x}$ exactly. For small differential augmentation, a first-order Taylor expansion of $\ln(\gamma_S(x)/\gamma_J(x))$ around $\hat{x}$ (holding wages fixed) yields

$$\hat{x}' - \hat{x} \approx \frac{\ln\left(\frac{1+\delta_J}{1+\delta_S}\right)}{\frac{d}{dx} \ln\left(\frac{\gamma_S(x)}{\gamma_J(x)}\right)\Big|_{x=\hat{x}}} \geq 0. \quad (26)$$

The numerator is increasing in $\delta_J - \delta_S$; the denominator—the steepness of comparative advantage at the boundary—is positive and finite by Assumption 1. By the implicit function theorem applied to (25), $\hat{x}'$ is a continuous function of (δ_J, δ_S) .

A.6.2 Senior Employment Decomposition

By symmetric reasoning, the change in senior labor demand under $\alpha < \hat{x}$ is

$$\Delta L_S^* = \underbrace{0}_{\text{displacement}} + \underbrace{-\int_{\hat{x}}^{\hat{x}'} l_S^{\text{pre}}(x) dx}_{\text{reallocation } (\leq 0)} + \underbrace{\int_{\hat{x}'}^1 [l_S^{\text{post}}(x) - l_S^{\text{pre}}(x)] dx}_{\text{productivity } (\leq 0)}. \quad (27)$$

Seniors face no displacement (no senior tasks are automated). The reallocation effect is weakly negative—seniors lose tasks near the boundary to newly-productive juniors. The productivity effect operates in the same direction as for juniors (see Appendix A.6.4), but

the within-task augmentation component is smaller when $\delta_S < \delta_J$; the cost-index channel, which is common to both worker types, is unchanged. The absence of displacement—the dominant channel for the junior decline—yields a muted net effect on senior labor demand.

A.6.3 Proof of Proposition 1

Proof. The change in junior labor demand is

$$\Delta L_J^* = L_J^{*\text{post}} - L_J^{*\text{pre}} = \int_{\alpha}^{\hat{x}'} l_J^{*\text{post}}(x) dx - \int_0^{\hat{x}} l_J^{*\text{pre}}(x) dx. \quad (28)$$

Splitting each integral at the shared endpoints α and $\hat{x}$:

$$\begin{aligned} \Delta L_J^* &= \left[\int_{\alpha}^{\hat{x}} l_J^{*\text{post}}(x) dx - \int_{\alpha}^{\hat{x}} l_J^{*\text{pre}}(x) dx \right] + \int_{\hat{x}}^{\hat{x}'} l_J^{*\text{post}}(x) dx - \int_0^{\alpha} l_J^{*\text{pre}}(x) dx \\ &= \underbrace{- \int_0^{\alpha} l_J^{*\text{pre}}(x) dx}_{\text{displacement}} + \underbrace{\int_{\alpha}^{\hat{x}} [l_J^{*\text{post}}(x) - l_J^{*\text{pre}}(x)] dx}_{\text{productivity}} + \underbrace{\int_{\hat{x}}^{\hat{x}'} l_J^{*\text{post}}(x) dx}_{\text{reallocation}}. \quad \square \end{aligned}$$

A.6.4 Sign of the Productivity Effect

We maintain the partial-equilibrium assumption that Y and wages (w_J, w_S) are fixed.

Under cost minimization with the CES aggregator (1), conditional demand for task x is $y^*(x) = Y \cdot (p(x)/P)^{-\sigma}$, where $p(x)$ is the unit cost of task x and

$$P = \left(\int_0^1 p(z)^{1-\sigma} dz \right)^{\frac{1}{1-\sigma}} \quad (29)$$

is the CES cost index (with the Cobb-Douglas limit $P = \exp\left(\int_0^1 \ln p(z) dz\right)$ when $\sigma = 1$). For a junior task $x \in (\alpha, \hat{x})$, the unit cost is $p(x) = w_J/\gamma_J(x)$ and output is $y(x) = \gamma_J(x) l_J(x)$, so the cost-minimizing labor allocation is

$$l_J^*(x) = Y \cdot P^{\sigma} \cdot w_J^{-\sigma} \cdot \gamma_J(x)^{\sigma-1}. \quad (30)$$

After GenAI adoption, the cost index changes from P^{pre} to P^{post} . On surviving junior